

Yuqing Wang

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<https://yzwangyuqing.github.io>

RESEARCH INTERESTS

deep learning theory, optimization, (stochastic) dynamical systems, sampling, computational mathematics

PROFESSIONAL EXPERIENCE

2025 - Postdoctoral Fellow

Department of Applied Mathematics and Statistics
Johns Hopkins University

Fall 2024 Research Fellow

Program: *Modern Paradigms in Generalization*, joint with *Special Year on Large Language Models and Transformers Part 1*

Simons Institute for the Theory of Computing, University of California, Berkeley

EDUCATION

Aug 2024 Ph.D. in Mathematics

Georgia Institute of Technology
Advisor: Prof. Molei Tao

June 2018 B.S. in Mathematics

Nankai University

PREPRINTS

Yuqing Wang, Youssef Marzouk, Molei Tao. *Quantifying general irreversible acceleration in SDEs for sampling, with an emphasis on state-dependent designs.*

Yuqing Wang, Ioannis G. Kevrekidis, Mikhail Belkin. *A Theoretical Analysis of Generalization Dynamics in Neural Networks under Gradient Descent with Weight Decay.* arXiv preprint arXiv:2609.07755

Yuqing Wang. *Convergence of Gradient Descent for General Neural Network Architectures Beyond the NTK Regime.* arXiv preprint arXiv:2606.23364

Shangding Gu, **Yuqing Wang**. *Rethinking Data Scaling: From Data Uniformity to Effective Learning Volume.*

PUBLICATIONS

Yuqing Wang, Ye He, Molei Tao. *Evaluating the design space of diffusion-based generative models*. Advances in Neural Information Processing Systems 37 (NeurIPS 2024).

Zhenghao Xu, **Yuqing Wang**, Tuo Zhao, Rachel Ward, Molei Tao. *Provable Acceleration of Nesterov’s Accelerated Gradient for Rectangular Matrix Factorization and Linear Neural Networks*. Advances in Neural Information Processing Systems 37 (NeurIPS 2024).

Yuqing Wang, Zhenghao Xu, Tuo Zhao, Molei Tao. *Good regularity creates large learning rate implicit biases: edge of stability, balancing, and catapult*. Journal of Machine Learning Research 26 (2025) 1-68 (JMLR).

Bo Yuan, Jiaojiao Fan, **Yuqing Wang**, Molei Tao, and Yongxin Chen. *Markov Chain Monte Carlo for Gaussian: A Linear Control Perspective*. IEEE Control Systems Letters, vol. 7, pp. 2173-2178, 2023, doi: 10.1109/LCSYS.2023.3285140.

Lingkai Kong, **Yuqing Wang**, and Molei Tao. *Momentum Stiefel Optimizer, with Applications to Suitably-Orthogonal Attention, and Optimal Transport*. International Conference on Learning Representations (ICLR 2023).

Yuqing Wang, Minshuo Chen, Tuo Zhao and Molei Tao. *Large Learning Rate Tames Homogeneity: Convergence and Balancing Effect*. International Conference on Learning Representations (ICLR 2022).

Kaixuan Huang*, **Yuqing Wang***, Molei Tao, and Tuo Zhao. *Why Do Deep Residual Networks Generalize Better than Deep Feedforward Networks?—A Neural Tangent Kernel Perspective*. Advances in neural information processing systems 33 (NeurIPS 2020). (*: equal contribution)

TALKS

Nov 2026 *A Theoretical Analysis of Generalization Dynamics in Neural Networks*, Algebraic, analytic and algorithmic approaches to generalization in machine (MS90), SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, Utah

Jan 2026 *Data uniformity improves training efficiency and more, with a convergence framework beyond the NTK regime*, Modern Paradigms in Generalization Reunion, Simons Institute, University of California, Berkeley

Jan 2026 *Data uniformity improves training efficiency and more, with a convergence framework beyond the NTK regime*, AMS Special Session on Algebraic Methods in Machine Learning and Optimization, Joint Mathematics Meetings (JMM 2026), Washington, D.C.

- Nov 2025 *Data uniformity improves training efficiency and more, with a convergence framework beyond the NTK regime*, Scientific Machine Learning with Robust Computation Minisymposium, SIAM New York-New Jersey-Pennsylvania Section Conference, Pennsylvania State University
- July 2025 *Theoretical implications of training and sampling diffusion models*, “Optimization Meets GenAI”, International Conference on Continuous Optimization (ICCOPT), University of Southern California
- May 2025 *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, Math Machine Learning seminar, Max Planck Institute for Mathematics in the Sciences + University of California, Los Angeles
- May 2025 *Theoretical implications of training and sampling diffusion models*, Dynamical Systems for Machine Learning (MS141), SIAM Conference on Applications of Dynamical Systems (DS25), Denver, Colorado
- Apr 2025 *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, Computational Mathematics + X (CMX) seminar, California Institute of Technology
- Apr 2025 *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, Applied Mathematics and Statistics (AMS) postdoc seminar, Johns Hopkins University
- Oct 2024 *The mechanism behind the implicit biases of large learning rates: edge of stability, balancing, and catapult*, SIAM Conference on Mathematics of Data Science (MDS24), October 21–25, 2024, Atlanta, Georgia
- Sep 2024 *How well does diffusion model generate?—A training and sampling combined quantification*, Emerging Generalization Settings Workshop, Simons Institute, University of California, Berkeley (joint with Molei Tao)
- Apr 2024 *What creates edge of stability, balancing, and catapult*, the Second Southeast Applied and Computational Math Student Workshop, Georgia Institute of Technology
- Dec 2023 *Quantitative acceleration of convergence to invariant distribution by irreversibility in diffusion processes*, PDE seminar, Georgia Institute of Technology
- Oct 2023 *What creates edge of stability, balancing, and catapult*, ACO student seminar, Georgia Institute of Technology
- May 2023 *Implicit bias of large learning rate*, SIAM Conference on Applications of Dynamical Systems (DS23), Machine Learning for Dynamical Systems & Dynamical Systems for Machine Learning - Part I of III (MS143), Portland, Oregon

HONORS

2026	<i>Rising Star (with selected talk)</i> , IOE-ISyE-MS&E Joint Rising Stars Workshop, University of Michigan
2026	<i>Rising Stars Award</i> , Conference on Parsimony and Learning (CPAL 2026), Tübingen, Germany
2025	<i>Sigma Xi Best Ph.D. Thesis Award</i> , Georgia Institute of Technology
2024	<i>Outstanding Teaching Assistant</i> , Georgia Institute of Technology
2023	<i>Rising Star in Data Science</i> , University of Chicago & University of California San Diego
2023	<i>Rising Star in Electrical Engineering and Computer Sciences (EECS)</i> , Georgia Institute of Technology
2023	<i>Top Graduate Student</i> , Georgia Institute of Technology
2022	<i>“Thank a teacher” Award</i> , Georgia Institute of Technology

SERVICE

Organizer:

Nov	2026	Minisymposium on “Algebraic, analytic and algorithmic approaches to generalization in machine”, SIAM Conference on Mathematics of Data Science (MDS26), Salt Lake City, Utah
May	2025	Minisymposium on “Dynamical Systems for Machine Learning”, SIAM Conference on Applications of Dynamical Systems (DS25), Denver, Colorado
2025-	2026	Applied Mathematics and Statistics postdoc seminar, Johns Hopkins University
Fall	2024	Modern Paradigms in Generalization/Large Language Models program seminar, Simons Institute, University of California, Berkeley
Fall	2024	Women+ mentoring sessions, Simons Institute, University of California, Berkeley
Oct	2024	Minisymposium on “The dynamical view of machine learning”, SIAM Conference on Mathematics of Data Science (MDS24), Atlanta, Georgia
Fall	2021	SIAM student seminar, Georgia Institute of Technology

Reviewer:

Conferences: NeurIPS, ICLR, ICML

Journals: Nature Communications, Journal of Machine Learning Research (JMLR)

Others:

Fall 2021 Vice President for SIAM Student Chapter, Georgia Institute of Technology

TEACHING

Instructor in the Department of Applied Mathematics and Statistics, Johns Hopkins University:

Fall 2025, Fall 2026
EN 553.432 Bayesian Statistics

Spring 2025, Spring 2026
EN 553.361 Introduction to Optimization I

Teaching Assistant in School of Mathematics, Georgia Institute of Technology:
teach 4 hours of studios per week

Fall 2018, Spring 2019, Fall 2019, Fall 2020, Spring 2021, Fall 2023
MATH 2552 Differential Equation

Spring 2020
MATH 2551 Multivariable Calculus

Summer 2019, Summer 2024
MATH 2550 Introduction to Multivariable Calculus

Head Teaching Assistant in the School of Mathematics, Georgia Institute of Technology:
design worksheets; teach 2 hours of studios per week

Spring 2022, Spring 2023
MATH 2552 Differential Equation